Python for Automation

Why Python for Automation?

Python is the most popular language for automation due to its simple syntax, extensive library support, and cross-platform compatibility.

File and Folder Automation

Python's os and shutil libraries allow you to automate file operations such as renaming, moving, copying, and deleting files and directories.

The pathlib library provides an object-oriented approach to file system paths, making file automation cleaner and easier to read.

Email Automation

Python's smtplib and email libraries allow you to send automated emails via SMTP servers like Gmail.

IMAP libraries allow you to read and scan incoming emails, enabling reply detection and inbox monitoring automation.

Web Scraping

BeautifulSoup and Scrapy are Python libraries used to extract data from websites automatically, enabling data collection pipelines.

Selenium allows browser automation for tasks like form filling, login workflows, and testing web applications.

Task Scheduling

The schedule library and APScheduler allow you to run Python scripts at specific times or intervals, enabling fully automated workflows.

On Linux and Mac, cron jobs can trigger Python scripts. On Windows, Task Scheduler serves the same purpose.